

FINAL PROJECT DOCUMENTATION

Desktop Application with Database Integration

TITLE PAGE

Include the following:

- Project Title
 - Student Name(s)
 - Course / Subject
 - Professor's Name
 - School Name
 - Date of Submission
-

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IMPORTANT: See below details for reference.

INTRODUCTION

Provide a short overview of your project:

- Purpose of the system
 - Description of the chosen company/enterprise
 - Importance of database systems in modern applications
 - Overview of what the project will accomplish
-

DELIVERABLE 1: REQUIREMENTS ANALYSIS AND DESIGN (45 MARKS)

1.1 Company / Enterprise Overview

- Name of the company (real)
- Type of business (e.g., Banking, Retail, Hospital, School System)
- Description of operations

Example:

The system is designed for a **Banking Management System** that handles customer accounts, transactions, and loan processing.

1.1 System, Data and Information Requirements

- **System Architecture**

System architecture is the conceptual model that defines the structure, behavior, and more views of a system. It is a formal description and representation of a system, organized in a way that supports reasoning about the structures and behaviors of the system. The architecture of a system reflects how the system is used and how it interacts with other systems and the outside world.

- Identify at least **7 entities**. Example:
 - Customer

- Account
- Transaction
- Employee
- Loan
- Branch
- Payment

For each entity, describe:

- What data is stored
 - Why it is important
-

1.3 Functions and Transactions

Explain system operations such as:

- Add new customer
- Open account
- Deposit / Withdraw money
- Process loan applications
- Generate reports

Clearly describe how each function interacts with the database.

1.4 Advantages of DBMS

Discuss benefits such as:

- Data consistency
 - Reduced redundancy
 - Improved security
 - Multi-user access
 - Data integrity
 - Backup and recovery
-

1.5 Entity Relationship Diagram (ERD) and Class Diagrams

Include:

- Entities
- Relationships
- Cardinality (1:1, 1:M, M:N)
- Primary Keys & Foreign Keys

Must follow **3rd Normal Form (3NF)**:

- No repeating groups

- No partial dependency
- No transitive dependency

1.6 Data Dictionary

Create a table format like this:

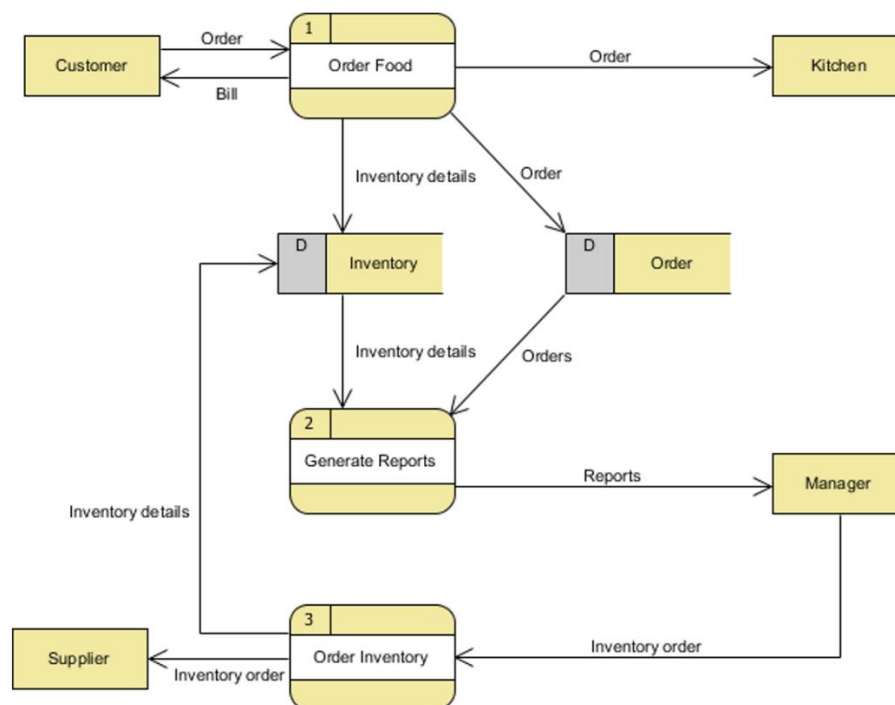
Table Name	Field Name	Data Type	PK/FK	Description	Constraints
------------	------------	-----------	-------	-------------	-------------

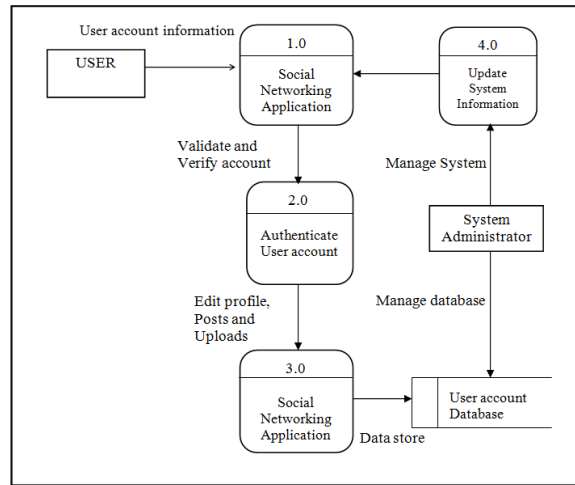
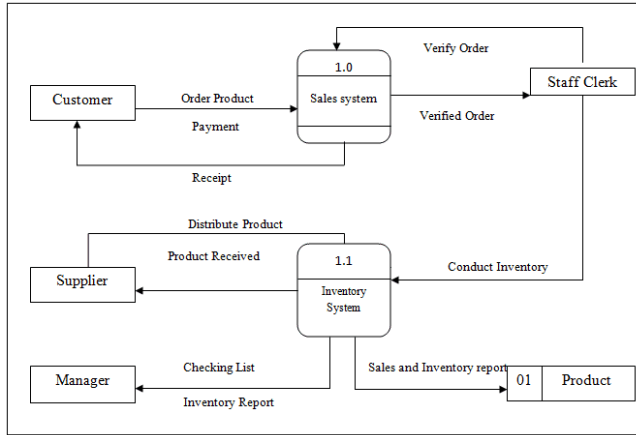
Include:

- All tables
- Attributes
- Keys
- Constraints / Business rules

1.7 Data Flow Diagram

- **A Data Flow Diagram (DFD)** is a visual modeling tool used to represent how data moves through a computer information system. It illustrates data inputs, processes, data stores, and outputs, making it easier to understand system functionality without focusing on implementation details.
- **Core Purpose:** DFDs help in system analysis, requirement gathering, and process optimization by showing how information flows between components. They are widely used in software engineering, business process modeling, and IT system design.





1.8 Program Design

- **Logic Formulation**
 - **Flowchart**

DELIVERABLE 2: DEVELOPMENT / IMPLEMENTATION (40 MARKS)

2.1 SQL Table Creation

Provide SQL scripts:

```

CREATE TABLE Customer (
    CustomerID SERIAL PRIMARY KEY,
    Name VARCHAR(100),
    Address TEXT,
    ContactNumber VARCHAR(15)
);

```

Include screenshots or output of created tables.

2.2 Sample Data (At least 20 records per table)

```

INSERT INTO Customer (Name, Address, ContactNumber)
VALUES
('Juan Dela Cruz', 'Manila', '09123456789'),
('Maria Santos', 'Quezon City', '09987654321');

```

2.3 SQL Queries (Minimum 7 Queries)

Must include:

- SELECT
- UPDATE
- DELETE
- WHERE
- AND / OR / NOT
- ORDER BY
- GROUP BY
- HAVING
- Aggregate Functions (COUNT, AVG, SUM)
- CASE Expression
- JOIN (2 or more tables)
- Subqueries (1–2)
- Implement the following DB Objects
 - INDEX
 - VIEW
 - STORED PROCEDURES
 - TRIGGERS
 - FUNCTIONS

Example:

```
SELECT Name, COUNT(AccountID)
FROM Customer
JOIN Account ON Customer.CustomerID = Account.CustomerID
GROUP BY Name;
```

2.4 Desktop Application (GUI + Database)

Describe your system:

- IDE, Programming Language and Database System used (NetBeans, Java, SQL Server, PostgreSQL, etc.)
- Features:
 - Add records
 - View records
 - Update/Delete
 - Search record
 - Report Generation

Include screenshots:

- **Main Menu / Dashboard**
 - **Login Page**
 - Forms (Insert, View, etc.)
 - Output display (Tables for View records)
 - Report / Data Visualization
-

DELIVERABLE 3: ASSESSMENT AND EVALUATION (15 MARKS)

Discussion

Address the following:

✓ Assumptions Made

- Example: “We assumed each customer can have multiple accounts...”

✓ System Effectiveness

- Does the database meet requirements?

✓ Transaction Support

- How well does the system handle operations?

✓ Challenges Encountered

- ERD design issues
- SQL errors
- GUI integration problems

✓ Solutions Applied

- Normalization fixes
- Query optimization
- Debugging strategies

CONCLUSION

Summarize:

- Project achievements
- Importance of database systems
- Learning outcomes

APPENDICES Section: Handwritten Reflection (Required, Bond Paper)

A. Students must submit:

- Minimum **15 sentences**
- Topic: **Eureka Moments, Learning, Reflections, and Insights**

Guide questions:

- What did you learn?
- What was difficult or challenging?
- What did you enjoy?
- How will you apply this in real life?
- What database concepts and requirements stood out to you, and why are they important for your future career as an IT professional?

B. Proof of Client Authenticity (Required)

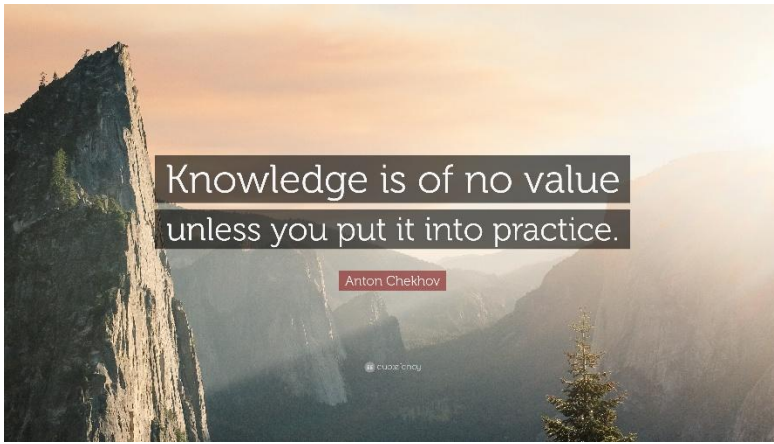
C. Others

- Full SQL scripts
- Additional screenshots - **including Group Collaborations / Meeting evidences like Pictures, Interview with the clients, F2F and Online Discussions**
- Extra diagrams
 - **Screenshots of your system** – Capture key parts of your application, including:
 - Log in page
 - Main dashboard/interface
 - All available modules/functionalities
 - Logout page/process

Ensure that screenshots are **clear, properly labeled, and arranged in sequence** to reflect the actual system flow.

FINAL NOTES

- Ensure your system is **functional and working**
- Follow **proper formatting and organization**
- **Group Presentation and Submission** of both:
 - Printed documentation
 - Working desktop application



Knowledge is of no value
unless you put it into practice.

Anton Chekhov

choose to try

#LearningByDoing #PracticeMakesPermanent